

1 Link-Level Precipitation Sensitivity of the UK Strategic 2 Road Network: An Empirical Framework Using Open 3 Data

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5 Abstract

As climate projections indicate more intense rainfall, understanding the precipitation resilience of road infrastructure is critical. Yet existing precipitation-speed estimates are reported as network-level averages, masking the link-level hydrological vulnerability needed for targeted drainage investment. This paper develops a three-stage framework (Estimate, Characterise, Map) to quantify, explain, and visualise link-level precipitation sensitivity on the UK Strategic Road Network. A Gamma generalised linear model with link-specific precipitation interactions, fitted to 5,050,333 weekday hourly speed observations across 460 monitored links in Cambridgeshire and Peterborough (2022–2024), recovers 201 link-specific sensitivity coefficients. By controlling for traffic volume, the model isolates the supply-side speed penalty, capturing the compound effect of rainfall intensity and unobserved drainage capacity; an additional millimetre per hour reduces speed by approximately 1.1% at the reference link. Corridor averages vary by a factor of five: high-standard dual carriageways (A11, A14) show minimal reductions (0.8–0.9% per mm/hr), while single-carriageway connectors approach 4%. A cross-sectional regression links these coefficients to observable road attributes; the number of lanes and road gradient emerge as the strongest correlates of resilience, consistent with crossfall design and gravitational runoff expectations. Crucially, the entire analysis relies solely on publicly available traffic and weather data and can be replicated by any stakeholder or researcher, circumventing the high costs and limited transferability of agent-based

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microsimulation. The coefficients are deployed via an open-source interactive platform that accepts user-specified rainfall intensities, enabling both current vulnerability diagnosis and forward-looking climate risk screening for adaptation prioritisation by any UK local authority with sensor coverage.

6 *Keywords:* precipitation sensitivity, hydrological vulnerability, precipitation
7 resilience, Gamma GLM, Strategic Road Network, drainage adaptation,
8 open-source platform

9 1. Introduction

10 Only 7% of the UK Strategic Road Network meets climate resilience stan-
11 dards established two decades ago (New Civil Engineer, 2026), yet this net-
12 work carries approximately one-third of all road traffic and two-thirds of freight
13 tonne-kilometres (National Highways, 2024). As climate projections indicate
14 more frequent and intense rainfall events across the United Kingdom (Depart-
15 ment for Transport, 2024; Sias et al., 2025), and 36% of the network already
16 exhibits significant susceptibility to flooding (Office of Rail and Road, 2024),
17 identifying which segments of this critical infrastructure are most vulnerable to
18 rainfall stress has become an urgent infrastructure adaptation challenge. Rain-
19 fall reduces traffic speeds, increases journey-time variability, and elevates crash
20 risk (Koetse and Rietveld, 2009; Hranac et al., 2006). The mechanisms are
21 both direct and indirect: surface water accumulation reduces tyre–road friction
22 and generates spray that impairs visibility, while localised pluvial flooding can
23 force lane closures or route diversions (Pregolato et al., 2017). The severity
24 of these effects depends not only on rainfall intensity but also on the road’s
25 capacity to shed water, a function of crossfall gradient, drainage system design,
26 and pavement surface condition.

27 Despite this policy urgency, most existing estimates of precipitation effects
28 on traffic speed are reported as network-wide or corridor-level averages. The
29 FHWA study documented free-flow speed reductions of up to 9% and speed-at-
30 capacity reductions of 8–14% during rain across three US metropolitan areas
31 (Hranac et al., 2006). Subsequent work has refined these estimates using radar-
32 based precipitation data (Becker et al., 2026), floating car data from ride-hailing

platforms (Bi et al., 2022), and urban road-section analysis (Cai et al., 2016). These aggregate benchmarks, however, mask what matters most for adaptation planning: individual road segments respond very differently to the same rainfall event. Knowing which segments are vulnerable, and by how much, is essential for targeting investment, yet link-level evidence is largely absent from the literature.

Research on road climate resilience varies not only in spatial scale but also in analytical approach. Simulation-based methods, particularly agent-based microsimulation (Huang et al., 2022; He et al., 2026) and coupled flood-traffic models (Pregolato et al., 2017; Farahmand et al., 2024), can capture counterfactual scenarios and model traffic dynamics under different weather conditions. However, building such models requires origin–destination matrices, detailed behavioural calibration, and often high-performance computing infrastructure. The development cost is substantial, and once calibrated for a specific city or region, the models are difficult to transfer elsewhere. As a result, these tools remain largely confined to well-resourced research groups and are not routinely available to the local transport authorities who most need practical climate adaptation evidence. Local authorities tasked with climate adaptation are thus caught in an evidence gap: aggregate traffic statistics are too coarse for targeted investment, while high-fidelity simulation tools are too resource-intensive for routine network-wide screening.

Three questions guide the analysis:

- (1) How large is the variation in precipitation sensitivity across individual road segments within a single regional network, and what spatial patterns emerge at the corridor level?
- (2) Which observable road characteristics (design standard, geometry, traffic load) are associated with greater or lesser hydrological vulnerability, and what engineering mechanisms (e.g., drainage design capacity, gravitational runoff) might explain these associations?
- (3) Can this evidence be produced at low cost using only publicly available data and standard computing resources, making it accessible to local

63 transport authorities?

64 To address these questions, we propose a three-stage framework (Estimate,
65 Characterise, Map). First, a Gamma GLM with link-specific precipitation in-
66 teractions is fitted to a large hourly panel to recover marginal speed penalties
67 for each segment, isolating supply-side effects by conditioning on traffic volume.
68 Second, these coefficients are linked to observable road attributes to generate en-
69 gineering hypotheses about the determinants of heterogeneous precipitation sen-
70 sitivity. Third, the results are spatially mapped and deployed as an open-source
71 platform that requires only public data and standard computing resources.

72 The analysis yields what is, to our knowledge, the first set of link-level precip-
73 itation sensitivity coefficients for the UK SRN. The estimated speed reduction
74 per millimetre of rainfall ranges from under 1% on the most resilient corridors
75 (A11, A14) to approximately 4% on the most sensitive (Cambourne Road),
76 with clear corridor-level spatial patterns (Question 1). The number of lanes
77 and road gradient emerge as the strongest correlates, consistent with the expect-
78 tation that drainage design standards and gravitational runoff capacity drive
79 precipitation resilience (Question 2). The entire analysis uses only publicly
80 available data (National Highways WebTRIS and Met Office CEDA Archive),
81 requires no microsimulation or high-performance computing, and is packaged
82 as an open-source platform that any local authority can deploy for their own
83 network (Question 3). This directly addresses the evidence gap identified by
84 the UK Department for Transport’s Rapid Evidence Assessment, which high-
85 lights the need for better understanding of climate change impacts on transport
86 infrastructure (Department for Transport, 2024).

87 **2. Literature Review**

88 *2.1. Precipitation Effects on Traffic Performance*

89 Research on rainfall and traffic performance has evolved along two distinct
90 axes: the empirical estimation of speed penalties and the development of re-
91 silience assessment tools. Koetse and Rietveld (2009) synthesised evidence

92 across transport modes and identified precipitation as the weather variable with
93 the most consistent negative effect on road performance. Hranac et al. (2006)
94 quantified the relationship for US highways, reporting free-flow speed reduc-
95 tions of up to 9% and speed-at-capacity reductions of 8–14% during rainfall,
96 estimates that remain the most widely cited benchmarks in the field.

97 Subsequent work extended these findings in scale and data quality. In Hong
98 Kong, Cai et al. (2016) showed that precipitation increases not only average de-
99 lays but also journey-time unreliability, though their city-level analysis could not
100 identify segment-specific effects. Across multiple Chinese cities, Bi et al. (2022)
101 found that weather-induced speed reductions vary by road type and time of day,
102 but pooled estimation precluded recovery of individual link coefficients. In the
103 largest-scale study to date, Becker et al. (2026) combined radar precipitation
104 with floating car data across 1.5 million German road sections and found that
105 high-speed, multi-lane roads experienced proportionally larger speed reductions
106 than lower-capacity roads. Even this study, however, reported segment-type
107 averages rather than individual link estimates.

108 Across this literature, precipitation effects are estimated at aggregate spatial
109 scales (network-wide, city-level, or at most corridor-level) (Koetse and Rietveld,
110 2009; Hranac et al., 2006; Bi et al., 2022). How individual segments respond
111 to the same rainfall remains unquantified. The closest attempt at segment-
112 level analysis is Gao et al. (2024), who clustered Harbin road links by resilience
113 patterns using environmental variables. Their approach, however, pursued a dif-
114 ferent objective: it classified links by resilience pattern rather than estimating
115 individual marginal coefficients via regression. The reliance on a binary speed
116 threshold, non-public data, and the absence of volume, temporal, and disrup-
117 tion controls limits direct comparison with the continuous regression estimates
118 produced here.

119 2.2. *Transport Climate Resilience: Methods and Gaps*

120 Transport climate resilience research spans qualitative risk reviews (Wang
121 et al., 2020) and quantitative resilience methodologies (Calvert and Snelder,

2018), as well as simulation-based approaches. Ganin et al. (2017) used topological analysis to demonstrate resilience–efficiency trade-offs in transport networks, while Bergantino et al. (2024) identified data availability and methodological accessibility as persistent barriers to widespread adoption of empirical resilience assessment.

Agent-based microsimulation and coupled flood-traffic models offer detailed disruption modelling but demand origin–destination matrices, calibrated behavioural parameters, and computing infrastructure beyond most local authorities (Huang et al., 2022; Pregolato et al., 2017). Coupling flood inundation (e.g., CityCAT) with traffic assignment (e.g., MATSim) adds further complexity (Pregolato et al., 2017; He et al., 2026; Farahmand et al., 2024). These tools suit scenario modelling in specific cities but are poorly suited to rapid screening across large networks.

The resulting gap has practical consequences. The UK Department for Transport’s Rapid Evidence Assessment identifies a need for better understanding of climate change impacts on road infrastructure and highlights that cost-benefit analyses of adaptation remain limited and methodologically challenging (Department for Transport, 2024). Investment frameworks such as the real options approach of Zare and Miller-Hooks (2025) depend on spatially disaggregated vulnerability evidence that the current literature does not provide.

2.3. Road Drainage Design and Surface Water Management

The traffic speed literature reviewed above treats road infrastructure as a homogeneous entity, estimating average precipitation effects without engaging with the physical processes that mediate them. In practice, the speed reduction observed during rainfall reflects a chain of hydrological and engineering interactions: precipitation falls on the road surface, accumulates as a water film whose depth depends on rainfall intensity, crossfall gradient, pavement porosity, and drainage system capacity, and this water film degrades tyre–road friction and generates spray that impairs visibility (Pregolato et al., 2017). The UK Design Manual for Roads and Bridges (DMRB) specifies drainage design standards

that vary by road class: motorways and high-speed dual carriageways are designed with steeper crossfall (typically 2.5%), wider carriageways that provide additional conveyance capacity, and drainage systems dimensioned for higher return-period storms than single-carriageway A-roads. Approximately 93% of the SRN’s drainage assets were built before the most recent climate change allowances were introduced, meaning that much of the network operates under design assumptions that no longer reflect current or projected rainfall intensities (New Civil Engineer, 2026).

This engineering context implies that the precipitation sensitivity of any given road segment is not solely a function of rainfall intensity; it reflects the *compound interaction* between the meteorological input and the segment’s physical capacity to manage surface water. Quantifying this compound effect at the link level requires the kind of spatially disaggregated evidence that the traffic speed literature has not yet produced.

Two gaps therefore stand out: precipitation–speed relationships are reported almost exclusively as spatial averages, obscuring which segments are most hydrologically vulnerable; and the tools that could produce link-level evidence are too resource-intensive for routine use. The present study addresses both by estimating a Gamma GLM with link-specific precipitation interactions on a large hourly panel, recovering individual sensitivity coefficients for 201 road segments using only public data and a standard laptop. An open-source platform extends the analysis to non-technical users in local transport authorities.

3. Data and Study Area

3.1. Study Area

The study covers the Strategic Road Network within and around Cambridgeshire and Peterborough, United Kingdom (Fig. 4): segments of the A1(M), A1, A11, A14, A14(M), A428, A47, M11, and Cambourne Road, totalling 460 links monitored by National Highways’ WebTRIS system. Road types range from three-lane motorways (A1(M), M11) to single-carriageway connec-

181 tors (Cambourne Road). This area was selected for the availability of collocated
182 traffic, weather, and disruption data, and because the region’s flat, low-lying ter-
183 rain amplifies surface water accumulation, making rainfall sensitivity a direct
184 concern for local transport planning.

185 3.2. Data Sources

186 The analysis integrates four data sources, all publicly available:

187 *Traffic data.* Hourly average speed (mph) and total traffic volume (ve-
188 hicles per hour) for each of the 460 links are obtained from National High-
189 ways’ Web-based Transport Information System (WebTRIS). The study period
190 spans January 2022 to December 2024. Observations are restricted to week-
191 days (Monday–Friday) to focus on commuting and commercial traffic patterns,
192 yielding 5,050,333 hourly records after quality filtering (removal of zero-speed
193 observations and sensor malfunctions).

194 *Weather data.* Hourly temperature ($^{\circ}\text{C}$) and precipitation (mm) are sourced
195 from the Met Office CEDA Archive. Weather stations in the study area (Cam-
196 bridge NIAB, Monks Wood, Wittering, Cambridge Botanic Garden) are matched
197 to road links using nearest-station assignment. A binary indicator for sub-zero
198 temperatures in the preceding 6 hours is constructed to capture frost and ice
199 effects.

200 *Disruption data.* Hourly indicators for traffic accidents, roadworks, and
201 special events are obtained from Alchera Technologies via the National Highways
202 data portal. These indicators isolate the precipitation effect from incident-
203 related speed variation.

204 *Calendar variables.* Binary indicators for school term, university term,
205 and bank holidays are constructed from official schedules published by Cam-
206 bridgeshire County Council and the University of Cambridge.

207 3.3. Descriptive Statistics

208 Table 1 presents summary statistics for the estimation sample. The mean
209 speed across all links and hours is 54.3 mph (s.d. = 15.8), reflecting the mix

210 of motorway and A-road segments. Hourly precipitation averages 0.12 mm but
 211 is heavily right-skewed (max = 28.4 mm/hr), consistent with the intermittent
 212 nature of rainfall. Approximately 12% of hours record non-zero precipitation.

Table 1: Descriptive statistics of key variables (weekday sample, 2022–2024, $N = 5,050,333$)

Variable	Mean	Std. Dev.	Min	Max
Average speed (mph)	54.3	15.8	2.1	85.0
Total volume (veh/hr)	412	389	0	4,856
Precipitation (mm/hr)	0.12	0.58	0.0	28.4
Temperature (°C)	11.2	6.1	−8.3	37.2
School term (0/1)	0.58	—	0	1
University term (0/1)	0.42	—	0	1
Accident (0/1)	0.02	—	0	1
Roadworks (0/1)	0.08	—	0	1

213 4. Methodology

214 Fig. 1 summarises the three-stage framework: estimation of link-specific
 215 precipitation coefficients via a Gamma GLM (Stage 1: Estimate), exploratory
 216 characterisation of which road attributes correlate with hydrological vulnera-
 217 bility (Stage 2: Characterise), and spatial mapping with platform deployment
 218 (Stage 3: Map).

219 4.1. Conceptual Framework

220 The analytical approach rests on a conceptual model of how precipitation
 221 translates into observed speed reductions (Fig. 2). A rainfall event deposits
 222 water on the road surface; the resulting water film depth depends on rainfall
 223 intensity and on the segment’s *supply-side* drainage capacity (crossfall gradient,
 224 pavement porosity, drainage system design). The water film degrades tyre–
 225 road friction and generates spray that impairs visibility, triggering behavioural
 226 responses from drivers (speed reduction, increased headway) and, in extreme

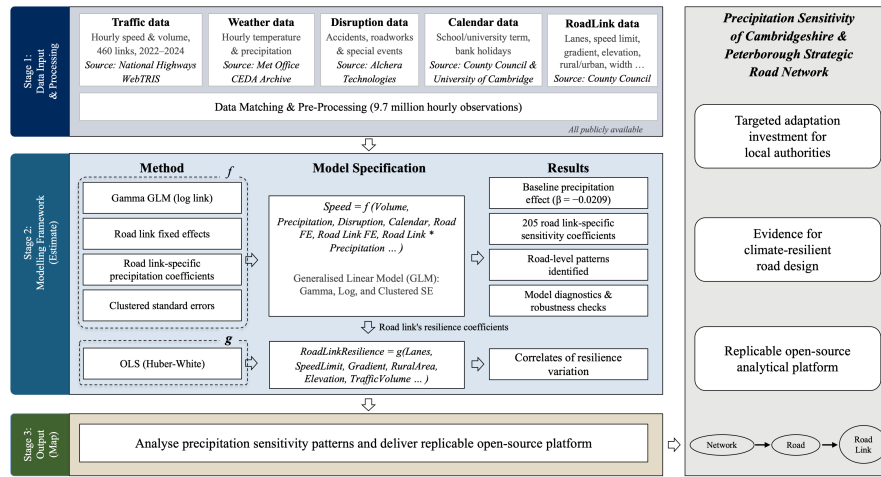


Figure 1: Analytical framework. Stage 1 (Estimate): a Gamma GLM with link-specific precipitation interaction terms yields 201 sensitivity coefficients. Stage 2 (Characterise): coefficients are linked to observable road characteristics (lanes, gradient, speed limit, daily volume) to identify correlates of hydrological vulnerability. Stage 3 (Map): coefficients are spatially mapped and deployed as an open-source platform.

cases, lane or road closures. These effects are captured by the WebTRIS sensors as hourly average speed reductions. Simultaneously, rainfall may reduce *demand-side* travel volumes as some drivers postpone or cancel trips. The Gamma GLM separates these channels by conditioning on observed traffic volume, isolating the supply-side speed effect, which is the parameter most relevant to infrastructure adaptation planning.

Conceptual Framework: Rainfall to Speed Reduction

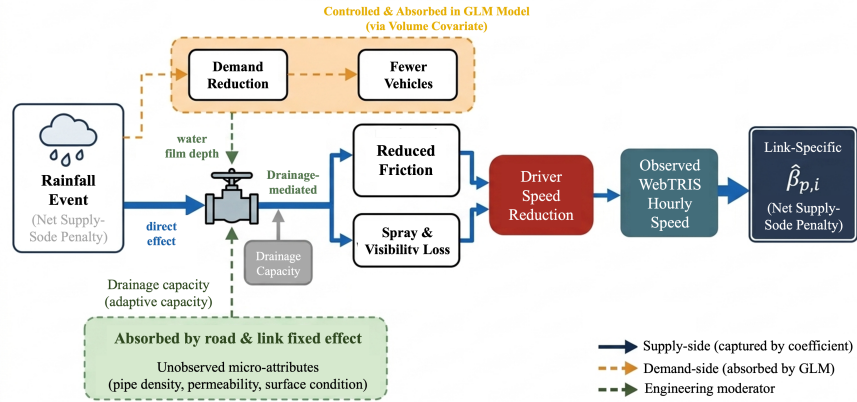


Figure 2: Conceptual framework linking rainfall to observed speed reductions. The precipitation coefficient estimated by the Gamma GLM captures the supply-side channel (solid arrows), conditional on observed traffic volume. The demand-side channel (dashed arrow) is absorbed by the volume control.

Because micro-scale engineering data (drainage pipe capacity, pavement permeability, crossfall angle) are not publicly available at the link level, the estimated precipitation coefficient for each segment reflects both the meteorological input and the segment's unobserved drainage capacity. Observable road characteristics are therefore used as proxies in the Characterise stage (Section 5.4).

4.2. Model Selection Rationale

Traffic performance under adverse weather has been measured by average speed (Hranac et al., 2006; Becker et al., 2026), travel time index (Bi et al.,

241 2022), journey-time reliability (Calvert and Snelder, 2018), and composite re-
 242 silience scores (Gao et al., 2024). Hourly average speed is used because it is the
 243 quantity directly observed by WebTRIS, requiring no free-flow benchmarks or
 244 disruption thresholds (Hranac et al., 2006). A continuous speed variable also
 245 suits Gamma GLM estimation (threshold-based scores require dichotomisation
 246 that discards information; Cai et al. 2016), and log-speed coefficients have a
 247 transparent percentage interpretation that aids communication to policy audi-
 248 ences.

249 Speed and volume are jointly determined through the speed–flow relation-
 250 ship, and rainfall may reduce both demand and speeds (Hranac et al., 2006;
 251 Becker et al., 2026). Omitting volume would conflate demand-side responses
 252 with the direct meteorological effect. Hourly volume is therefore included as a
 253 control (Section 4.4), following Becker et al. (2026) and Cai et al. (2016), so
 254 that the precipitation coefficient captures the speed effect of rainfall *conditional*
 255 *on observed volume*.

256 The distributional family and link function are chosen to address three prop-
 257 erties of the speed data that violate standard OLS assumptions. First, hourly
 258 average speed is strictly positive by construction. Second, the distribution is
 259 right-skewed (Table 1: mean 54.3 mph, s.d. 15.8). Third, variance increases
 260 with the conditional mean, as high-speed motorway segments exhibit greater
 261 absolute speed variation than low-speed local links, a pattern consistent with
 262 fundamental traffic flow theory, in which free-flowing high-speed segments (e.g.,
 263 60–70 mph) possess inherently greater absolute speed variation than congested
 264 urban links (e.g., 10–20 mph), making heteroskedasticity an expected physi-
 265 cal property rather than merely a statistical nuisance. These features, het-
 266 eroskedasticity in particular, make OLS inefficient and can bias standard errors
 267 (McCullagh and Nelder, 1989). Wan and Huang (2024) applied a Gamma GLM
 268 with a log link to model corridor-level journey time on the Cambridge road net-
 269 work, exploiting the same distributional properties. The present study extends
 270 this approach to the road link level, estimating segment-specific precipitation
 271 sensitivity coefficients rather than corridor averages.

272 OLS on untransformed speed permits negative predictions and imposes a
 273 constant absolute rainfall effect regardless of baseline speed. Log-OLS avoids
 274 negative predictions but models the conditional geometric mean; retransforma-
 275 tion introduces bias under heteroskedasticity (McCullagh and Nelder, 1989).

276 The Gamma GLM with a log link resolves all three problems simultaneously:
 277 the Gamma distribution is defined on the positive reals (addressing the positivity
 278 constraint), its variance function is proportional to the square of the mean
 279 (accommodating the mean-variance relationship), and the log link ensures that
 280 covariates operate multiplicatively, so that each millimetre of rain produces a
 281 proportional rather than absolute speed reduction.

282 Two alternative specifications were also tested: an inverse Gaussian GLM
 283 (sharper mean-variance relationship; failed to converge with the full interaction
 284 specification) and a log-normal model (OLS on log-speed; similar precipitation
 285 coefficients but subject to the retransformation bias noted above). The Gamma
 286 specification yielded the lowest AIC and was retained. Deviance residual diag-
 287 nostics are described in Section 4.6.

288 4.3. Model Specification

289 The model is specified as:

$$\log(\mathbb{E}[\text{speed}_{it}]) = \beta_0 + \beta_v \cdot \text{volume}_{it} + \sum_k \beta_k \cdot X_{kit} + \alpha_r + \gamma_i + \beta_p \cdot \text{precip}_{it} + \delta_i \cdot \text{precip}_{it} \quad (1)$$

290 where i indexes road links ($i = 1, \dots, 460$), r indexes the road (highway) to
 291 which link i belongs, and t indexes hours. The terms are structured in three
 292 layers:

- 293 • α_r : *road fixed effects*, one dummy per named highway (A1, A11, A14,
 294 A1(M), A428, A47, Cambourne Road, M11), absorbing road-level time-
 295 invariant characteristics such as design standard, typical maintenance
 296 regime, and corridor-level traffic composition.
- 297 • γ_i : *road link fixed effects*, one dummy per sensor-to-sensor segment, ab-
 298 sorbing all remaining link-specific time-invariant factors (localised drainage,

gradient, pavement condition) beyond what the road-level dummies capture. Because each link maps to exactly one road, γ_i nests α_r ; the road-level dummies are retained for expositional clarity but are not separately identified.

- $\delta_i \cdot \text{precip}_{it}$: *road link $\times$ precipitation interaction terms*, the parameters of primary interest. Each δ_i captures how the precipitation sensitivity of link i deviates from the baseline effect β_p . The total precipitation sensitivity for link i is $\beta_p + \delta_i$.

The remaining terms are: β_v , the volume–speed coefficient; X_{kit} , a vector of control variables (temporal fixed effects, calendar indicators, temperature, and disruption events); and β_p , the baseline precipitation effect at the reference link. The 201 composite coefficients ($\beta_p + \delta_i$) constitute the primary output of the estimation.

4.4. Control Variable Strategy

Each control absorbs a source of speed variation that could confound the precipitation estimate:

- *Traffic volume*: total hourly vehicle count. Speed and volume are jointly determined through the speed–flow relationship (Hranac et al., 2006). Rainfall may also independently reduce travel demand, as some drivers postpone or cancel trips during adverse weather (Koetse and Rietveld, 2009; Becker et al., 2026). Omitting volume would therefore conflate two channels: (a) demand-side reductions (fewer vehicles on the road, which may actually *increase* observed speeds) and (b) supply-side speed reductions (slower driving under wet conditions). Including volume as a control isolates channel (b), the direct meteorological effect, which is the parameter of policy interest. As a robustness check, re-estimating the model excluding volume shows that the precipitation coefficients remain qualitatively similar in sign and pattern, though slightly attenuated (Section 6.6).
- *Temporal fixed effects*: indicators for month, year, hour of day, and day of week. Precipitation is seasonal and diurnal; without temporal controls,

the precipitation coefficient would absorb systematic differences in speed between, for example, winter and summer that are unrelated to contemporaneous rainfall.

- *Calendar controls*: school term, university term, and bank holiday indicators. Term-time traffic patterns differ markedly from holiday patterns in a university city; since term dates correlate with season, omitting these controls would introduce seasonal confounding.
- *Temperature*: continuous temperature and a binary indicator for sub-zero temperatures in the preceding 6 hours. Cold weather affects both rainfall likelihood and road surface conditions; without temperature controls, the precipitation coefficient would partly capture frost and ice effects.
- *Disruption events*: binary indicators for accidents, roadworks, and special events. Rainfall increases accident risk, so some observed speed reductions during rain are caused by crash-related congestion rather than the direct effect of precipitation on driving conditions. Including disruption indicators isolates the direct meteorological channel.

Highway corridor dummies (α_r in Equation 1) absorb time-invariant corridor-level characteristics such as road design, speed limit, and surface quality. Link fixed effects (γ_i) absorb remaining link-specific time-invariant factors beyond what the corridor dummies capture.

4.5. Estimation and Inference

The model is estimated in Stata 19.5 using the `glm` command with `family(gamma)` `link(log)` specification. Standard errors are clustered at the link level to account for within-link serial correlation in hourly observations.

Inference therefore rests on 201 clusters rather than 5,050,333 observations. With 201 clusters and 201 link-specific interaction terms, the cluster-to-parameter ratio is adequate for asymptotic reliability, though effective degrees of freedom for individual interaction terms are modest.

The dataset contains 460 links monitored by WebTRIS, of which 213 have sufficient hourly observations (minimum 1,000 per link) and non-missing precip-

359 itation data to support stable estimation of link-specific interaction terms. After
 360 excluding links with collinear or near-zero-variance precipitation series, 200 link-
 361 specific interaction coefficients (δ_i) are estimated, with one link serving as the
 362 reference category ($\delta_{\text{ref}} = 0$). Together with the baseline precipitation effect β_p ,
 363 these yield 201 composite sensitivity estimates ($\beta_p + \delta_i$). The remaining 259
 364 links are retained in the model (absorbed by link fixed effects and contributing
 365 to the estimation of shared parameters) but do not receive individual precipita-
 366 tion interaction terms; they are predominantly short connector segments with
 367 low observation counts. Relaxing the threshold to 500 observations yields 218
 368 coefficients with noisier tail estimates but similar corridor-level patterns.

369 4.6. Model Diagnostics

370 Model adequacy was assessed through three checks (Appendix Appendix
 371 B). First, deviance residuals plotted against fitted values show no systematic
 372 curvature, supporting the Gamma distributional assumption and log-link spec-
 373 ification. Second, a Q–Q plot of deviance residuals shows approximate normal-
 374 ity in the central quantiles, with modest departures in the tails attributable
 375 to extreme-weather observations. Third, the correlation between observed and
 376 model-predicted speeds is 0.85. These diagnostics, together with the robustness
 377 checks in Section 6.5, support the chosen specification.

378 4.7. Interpretation of Coefficients

379 Under the log link, each coefficient represents an approximate proportional
 380 change in expected speed. The reference link (link 146, an A14 segment near
 381 the median of the sensitivity distribution) is defined by setting $\delta_{\text{ref}} = 0$; the
 382 choice of reference link does not affect the total sensitivity estimates ($\beta_p + \delta_i$)
 383 for any link. The baseline precipitation coefficient β_{precip} is interpreted as the
 384 percentage speed reduction per additional millimetre of hourly rainfall at the
 385 reference link, holding all controls constant. The link-specific interaction δ_i
 386 captures the *deviation* from this baseline for link i . Links with $\delta_i > 0$ (less
 387 negative total effect) are relatively more resilient to precipitation; links with
 388 $\delta_i < 0$ (more negative total effect) are relatively more sensitive.

389 4.8. Stage 2: Road Characteristic Regression

390 The second stage examines which observable road characteristics are associ-
 391 ated with greater or lesser precipitation sensitivity. The 201 composite coeffi-
 392 cients $\hat{\beta}_{p,i} = \beta_p + \delta_i$ estimated in Stage 1 serve as the dependent variable in a
 393 cross-sectional OLS regression:

$$\begin{aligned} \hat{\beta}_{p,i} = & \alpha + \gamma_1 \text{Lanes}_i + \gamma_2 \text{SpeedLimit}_i + \gamma_3 \text{DailyFlow}_i + \gamma_4 \text{Length}_i \\ & + \gamma_5 \text{Gradient}_i + \gamma_6 \text{Elevation}_i + \gamma_7 \text{Rural}_i + \varepsilon_i \quad (2) \end{aligned}$$

394 where $\hat{\beta}_{p,i}$ is the total precipitation sensitivity coefficient for link i (more neg-
 395 ative values indicate greater sensitivity); Lanes_i is the number of carriageway
 396 lanes; SpeedLimit_i is the posted speed limit (mph); DailyFlow_i is the average
 397 daily traffic volume (vehicles per day); Length_i is the segment length (metres);
 398 Gradient_i is the mean longitudinal gradient (%); Elevation_i is the mean eleva-
 399 tion above sea level (metres); and Rural_i is a binary indicator for rural road
 400 classification. Because more negative $\hat{\beta}_{p,i}$ values denote greater sensitivity, a
 401 positive γ_k indicates that the corresponding characteristic is associated with
 402 less sensitivity (greater precipitation resilience).

403 Heteroskedasticity-consistent standard errors (robust) are used throughout,
 404 as the Stage 1 coefficients are estimated with varying precision across links.
 405 Variance inflation factors (VIF) are examined to assess multicollinearity: all
 406 VIFs fall below the conventional threshold of 5 (maximum VIF = 2.67 for daily
 407 flow; mean VIF = 1.59), indicating that multicollinearity does not compromise
 408 coefficient stability.

409 Two methodological caveats apply. First, this is an exploratory cross-sectional
 410 analysis ($N = 201$) intended to identify correlates and generate engineering hy-
 411 potheses rather than to establish causal relationships. Observable character-
 412 istics may proxy for unobserved design features (for example, lane count may
 413 capture the bundle of DMRB drainage standards that accompany higher road
 414 classifications), and the coefficients γ_k should be interpreted as associations,

not causal effects. Second, the dependent variable $\hat{\beta}_{p,i}$ is itself estimated from Stage 1, introducing a generated-regressors problem: the OLS standard errors in Equation 2 do not account for the sampling uncertainty in $\hat{\beta}_{p,i}$. The HC1 estimator partially addresses heteroskedasticity arising from differential first-stage precision, but a fully efficient two-step variance correction would require incorporating the Stage 1 variance-covariance matrix. Given the large first-stage sample ($N > 5$ million observations) and the exploratory intent, this refinement is deferred to future work.

4.9. Stage 3: Spatial Mapping and Platform Deployment

The final stage maps the 201 estimated coefficients to their geographic coordinates and deploys the results through an open-source interactive platform. Each link’s total precipitation sensitivity ($\hat{\beta}_{p,i}$) is colour-coded on a map of the study area, enabling visual identification of corridor-level vulnerability patterns. Scenario-specific speed losses can be computed as:

$$Predicted_Loss_i = 1 - e^{\hat{\beta}_{p,i} \cdot R} \quad (3)$$

where R is a scenario rainfall intensity (mm/hr). This formula translates per-millimetre marginal effects into absolute percentage speed reductions under specific rainfall scenarios, providing transport planners with directly interpretable metrics for drainage investment prioritisation. The platform, built in Streamlit with Folium map layers, allows users to explore individual link coefficients, view corridor-level summary statistics, and upload their own WebTRIS and weather data to replicate the analysis for any UK road network with sensor coverage.

5. Results

5.1. Gamma GLM Estimation

With the full specification, the Gamma GLM converges on 5,050,333 weekday observations, 201 link fixed effects, and 201 link-specific precipitation interactions. Selected coefficients appear in Table 2.

Table 2: Gamma GLM results: selected coefficients (dependent variable: average speed, mph)

Variable	Coefficient	Std. Error	z-value
Precipitation	−0.0111***	0.0002	−53.3
Total volume	−0.00002***	0.000008	−3.2
Temperature	0.00032***	0.0001	3.20
Temp. below 0 (past 6h)	−0.0073***	0.0010	−7.00
School term	−0.0051***	0.0009	−5.82
University term	−0.0036***	0.0008	−4.83
Bank holiday	0.0374***	0.0037	10.12
Event	−0.0236***	0.0029	−8.14
Roadworks	−0.0102***	0.0023	−4.41
Accident	−0.0127***	0.0019	−6.58
<i>Model specification</i>			
Observations	5,050,333		
Clusters (links)	201		
Link fixed effects	200		
Link × precip. interactions	200 (ref: link 146)		
Family / link	Gamma / log		
AIC	10.111		
Deviance	99,411		

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Robust standard errors clustered at the link level.

Additional controls: month, year, hour-of-day, day-of-week FE; highway dummies.

441 The baseline precipitation coefficient is -0.0111 ($p < 0.001$): each additional
442 millimetre of hourly rainfall reduces expected speed by about 1.1% at the refer-
443 ence link. Because the model includes binary indicators for accidents and road-
444 works as control variables, this coefficient captures the residual speed penalty
445 after removing disruption-induced congestion, isolating the direct supply-side
446 effect of degraded driving conditions (reduced tyre-road friction and impaired

visibility from surface water) rather than secondary incident-related queueing. This is a marginal per-millimetre effect and is not directly comparable to the aggregate speed-at-capacity reductions of 8–14% reported for sustained rain events; see Section 6.2 for a detailed comparison.

All control variables are significant with expected signs (Table 2). Hour-of-day and month fixed effects are jointly significant, absorbing the expected diurnal and seasonal speed variation. Model diagnostics (Appendix Appendix B) confirm adequate specification: deviance residuals show no systematic pattern, and the correlation between observed and predicted speeds is 0.85.

5.2. Link-Level Heterogeneity

The 201 link-specific coefficients vary widely. Fig. 3 presents the distribution of total precipitation effects ($\beta_{precip} + \delta_i$).

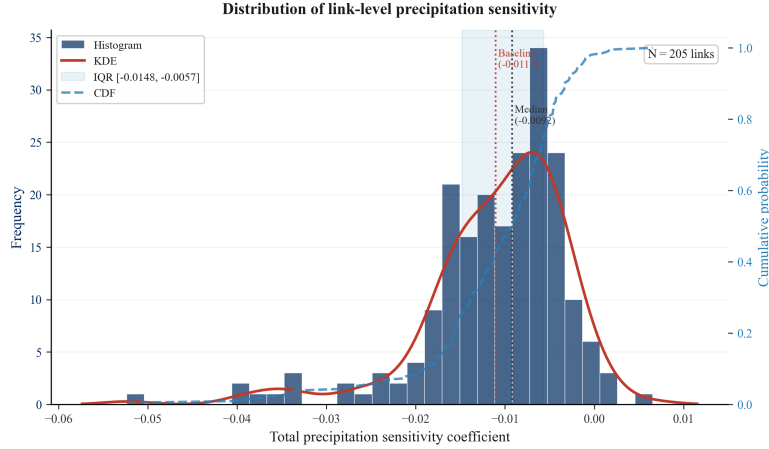


Figure 3: Distribution of link-level total precipitation sensitivity coefficients ($N = 201$). Navy bars: histogram; red curve: kernel density estimate; blue dashed curve: cumulative distribution function (right axis). The vertical dashed line marks the baseline coefficient (-0.0111); the shaded band indicates the interquartile range (-0.012 to -0.005).

Coefficients range from -0.042 (most sensitive) to $+0.019$ (least sensitive). The majority (189 of 201) are negative, indicating that precipitation reduces speed on most links. Twelve links show small positive coefficients, which may

462 reflect localised route diversion during wet conditions, where traffic displaced
 463 from highly sensitive parallel corridors shifts to these alternative links without
 464 exceeding their congestion thresholds, or statistical noise for coefficients near
 465 zero. The interquartile range (-0.012 to -0.005) corresponds to speed reduc-
 466 tions of 0.5–1.2% per millimetre.

467 5.3. Spatial Patterns

468 Mapping the coefficients (Fig. 4) indicates clear corridor-level patterns:

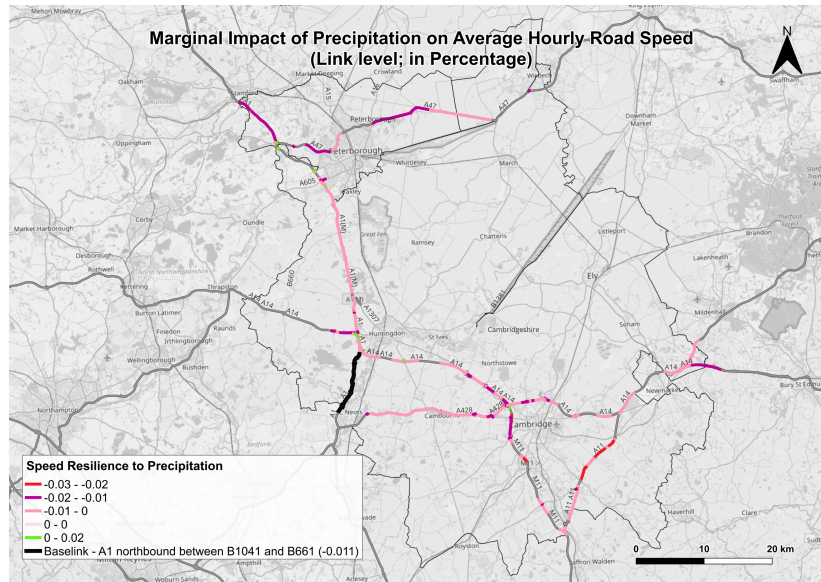


Figure 4: Spatial distribution of link-level precipitation sensitivity on the Cambridgeshire and Peterborough SRN. Colour scale ranges from green (resilient, coefficient near zero) to red (sensitive, large negative coefficient).

- 469 • *A11 and A14*: the least sensitive corridors, averaging -0.0075 and -0.0087
 470 respectively, both well above the network mean. Within-corridor variation
 471 on the A14 (s.d. = 0.0054) is moderate.
- 472 • *A1(M) and M11*: near the network average, at -0.0095 and -0.0124
 473 respectively.
- 474 • *Cambourne Road*: the most sensitive corridor, with coefficients near -0.039
 475 (3.9% speed reduction per mm/hr). Both links serve a rapidly growing

residential development.

- *A14(M)*: despite having the highest average gradient (0.7%) among all corridors, A14(M) exhibits above-average sensitivity (-0.015). With only 8 links and 2.0 average lanes, the limited lane count and small sample size likely outweigh the gradient advantage; this illustrates that no single road attribute determines sensitivity in isolation.
- *A1*: the largest within-corridor variation (s.d. = 0.013), with individual links ranging from -0.042 to $+0.019$.

Table 3 summarises the corridor-level patterns.

Table 3: Mean precipitation sensitivity and road characteristics by highway corridor

Corridor	N	Mean coeff.	Std. Dev.	Lanes (avg.)	Speed limit (mph, avg.)	Gradient (%, avg.)	Interpretation
A11	13	-0.0075	0.0025	2.0	70	0.3	Most resilient
A14	72	-0.0087	0.0054	2.5	70	0.2	Resilient
A1(M)	28	-0.0095	0.0062	2.9	70	0.5	Resilient
A428	13	-0.0122	0.0060	1.7	67	0.3	Near baseline
A47	10	-0.0123	0.0071	1.6	64	0.1	Near baseline
M11	29	-0.0124	0.0064	2.1	70	0.1	Near baseline
A1	30	-0.0137	0.0126	2.2	67	0.3	Sensitive
A14(M)	8	-0.0149	0.0111	2.0	70	0.7	Sensitive
Cambourne Rd	2	-0.0392	0.0004	1.0 [†]	40 [†]	— [†]	Most sensitive
All links	201	-0.0111	0.0081	2.3	69	0.3	—

[†]Cambourne Road is a local authority road not in the DfT database; lane count and speed limit are from field observation. Gradient data are unavailable for this corridor.

Corridor averages differ by a factor of approximately 5 (Cambourne Road at -0.039 versus A11 at -0.008). Within corridors the spread is wider still: the most sensitive single link (-0.042) is over five times the A11 corridor average.

Fig. 5 presents the within-corridor distributions as box plots, confirming that corridors differ not only in their central tendency but also in their inter-

nal dispersion. The A1 corridor exhibits the largest within-corridor variation (s.d. = 0.013). Cambourne Road, by contrast, has negligible internal variation (s.d. = 0.0004), consistent with its homogeneous design.

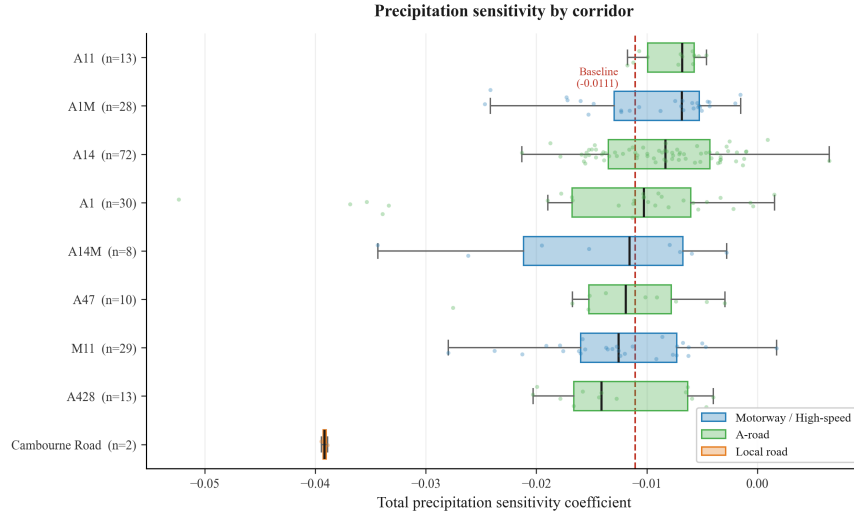


Figure 5: Distribution of total precipitation sensitivity coefficients by highway corridor. Boxes show interquartile range; whiskers extend to $1.5 \times \text{IQR}$. Individual links are overlaid as jittered points. The dashed line marks the baseline coefficient (-0.0111). Corridors are sorted by median sensitivity; blue = motorway/high-speed, green = A-road, orange = local road.

5.4. Correlates of Precipitation Sensitivity: Road Characteristic Analysis (Stage 2)

The second stage of the framework regresses the estimated sensitivity coefficients on observable road characteristics (Equation 2), examining which physical and operational attributes are associated with greater or lesser hydrological vulnerability. Two of the 201 links (both Cambourne Road segments) are excluded due to missing gradient data, yielding $N = 199$ for this cross-sectional regression. As noted in Section 4.8, this analysis is exploratory, intended to generate hypotheses about the engineering mechanisms behind the observed heterogeneity rather than to establish causal relationships, but its findings have direct implications for adaptation planning.

503 Seven road-level attributes are examined: number of lanes, speed limit, mean
 504 daily traffic volume, segment length, gradient, elevation, and rural/urban clas-
 505 sification. Among these, four show significant associations with precipitation
 506 sensitivity (Fig. 6):

- 507 • *Number of lanes* emerges as the strongest observable correlate of pre-
 508 cipitation resilience: links with more lanes are systematically less sensi-
 509 tive to precipitation. This association is consistent with the corridor-level
 510 finding that dual-carriageway and motorway segments outperform single-
 511 carriageway links, but the engineering interpretation extends beyond lane
 512 count per se. Multi-lane roads are typically built to higher design stan-
 513 dards specified in the Design Manual for Roads and Bridges (DMRB),
 514 including steeper crossfall gradients (typically 2.5% on motorways ver-
 515 sus 1.5–2.0% on single carriageways) that accelerate surface water runoff,
 516 larger-capacity drainage systems dimensioned for higher design flows, and
 517 wider hard shoulders that provide additional buffer capacity during in-
 518 tense rainfall. The lane-count variable therefore likely acts as a proxy
 519 for a bundle of unobserved drainage and geometric design characteristics
 520 that jointly determine a segment’s capacity to shed surface water. An al-
 521 ternative explanation is that multi-lane roads simply have higher baseline
 522 speeds, providing greater headroom for proportional speed reduction with-
 523 out the same operational consequence. However, if this speed-baseline ef-
 524 fect were the sole driver, the lane-count coefficient should lose significance
 525 once speed limit is controlled; the fact that lane count retains a strong
 526 independent association ($t = 5.41$) after conditioning on speed limit sug-
 527 gests that the drainage-design interpretation carries explanatory weight
 528 beyond the speed-baseline channel alone.
- 529 • *Gradient* shows a positive association with resilience: segments with steeper
 530 longitudinal gradients are less sensitive to rainfall. The study area lies
 531 within the flat, low-lying Cambridgeshire fenlands, where natural grav-
 532 ity drainage is limited. On near-flat segments, surface water accumu-
 533 lates more readily, reducing tyre–road friction and visibility through spray.

Road segments with even modest longitudinal gradients benefit from enhanced gravitational runoff, reducing the depth and duration of surface water films. This finding is consistent with the hydraulic engineering principle that drainage flow velocity (and hence clearance rate) increases with gradient.

- *Speed limit* is positively associated with resilience, though this variable is highly collinear with lane count (higher speed limits are assigned to roads with higher design standards). After accounting for lane count, the independent contribution of speed limit is modest, suggesting that both variables capture the same underlying design-quality dimension.

Four further variables complete the specification:

- *Mean daily traffic volume* is significantly associated with greater sensitivity ($t = -4.04$), likely reflecting that higher-flow links are more prone to congestion–weather compounding: when a road already operates near capacity, even a small rainfall-induced speed reduction triggers disproportionate delays through the non-linear speed–flow relationship.
- *Segment length* shows no significant association with precipitation sensitivity ($t = 0.23$), indicating that the physical length of a monitored segment does not systematically predict its hydrological vulnerability once other road characteristics are controlled.
- *Elevation and rural/urban classification* show no significant associations ($t = -0.48$ and $t = 0.95$, respectively), likely because these categories are too coarse to capture the micro-scale drainage and geometric features that determine surface water behaviour.

Table 4 reports the Stage 2 regression results. The dependent variable is the estimated total precipitation sensitivity coefficient ($\hat{\beta}_{p,i}$) for each road link.

In practical terms, the coefficients imply that, all else equal, adding one lane to a road is associated with a 0.53 percentage-point reduction in the speed penalty per mm/hr of rainfall, while a one percentage-point increase in longitudinal gradient is associated with a 0.29 percentage-point reduction. These magnitudes are directly relevant to engineering decisions: for a segment cur-

Table 4: Stage 2: road characteristic regression (dependent variable: precipitation sensitivity coefficient, $\hat{\beta}_{p,i}$)

Variable	Coefficient	Std. Error	<i>p</i> -value
Number of lanes	0.00529***	0.000978	<0.001
Speed limit (mph)	0.00117***	0.000420	0.006
Daily flow (vehicles)	-3.23×10^{-7} ***	8.00×10^{-8}	<0.001
Segment length (m)	5.90×10^{-8}	2.58×10^{-7}	0.819
Gradient (%)	0.00287***	0.000997	0.004
Elevation (m)	-1.75×10^{-5}	3.64×10^{-5}	0.632
Rural (binary)	0.000941	0.000990	0.343
Constant	-0.0853***	0.0285	0.003
<i>Model specification</i>			
Observations	199		
R^2	0.216		
Adjusted R^2	0.187		
F -statistic	8.13		
Estimation	OLS with robust SE		

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Heteroskedasticity-robust standard errors.

Dependent variable is the Stage 1 estimated coefficient; generated-regressors caveat applies.

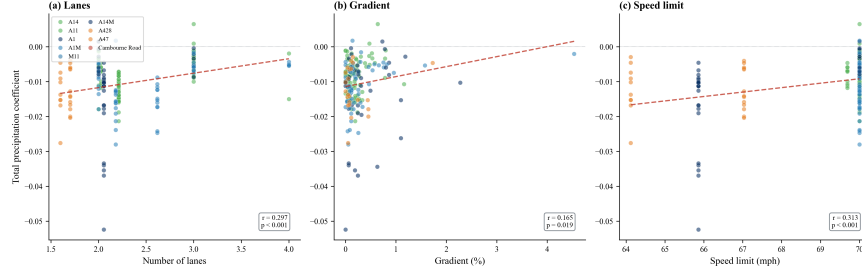


Figure 6: Road characteristics and precipitation sensitivity. Each point represents one of 201 road links. Left: number of lanes versus total precipitation coefficient; centre: gradient versus coefficient; right: speed limit versus coefficient. Dashed lines show linear fit. Links with more lanes, steeper gradients, and higher speed limits tend to exhibit lower (less negative) sensitivity, indicating greater precipitation resilience.

rently experiencing a 2% speed penalty per mm/hr, upgrading from single to dual carriageway would, on the basis of this association, halve the estimated rainfall impact.

Observable road characteristics capture a meaningful but incomplete share of the cross-link variation in precipitation sensitivity. Unobserved factors, particularly drainage system capacity, pavement porosity, and maintenance condition, may partially explain the residual heterogeneity. Crucially, this incompleteness has a practical implication: the high-sensitivity segments that cannot be explained by observable characteristics are precisely those where hidden drainage deficiencies or deteriorated pavement conditions are most likely, making them priority candidates for targeted engineering inspection. These exploratory associations are subject to the generated-regressors caveat, as the dependent variable (link-level sensitivity) is itself estimated with varying precision across links.

6. Discussion

6.1. Interpretation of Key Findings

Individual link-level precipitation sensitivity varies by a factor of approximately 20 (from -0.002 to -0.042 among the 189 negatively-signed links),

582 compared with a factor of approximately 5 across corridor-level averages (Sec-
583 tion 5.3), revealing substantial heterogeneity in the hydrological vulnerability
584 of road infrastructure at both scales. Aggregate estimates, such as the 8–14%
585 speed-at-capacity reductions reported by Hranac et al. (2006), are insufficient
586 for targeted adaptation at the local level. A local authority allocating drainage
587 investment on network-average vulnerability would systematically underinvest
588 in the most sensitive segments and overinvest in resilient ones.

589 The pattern is spatially coherent: corridors such as the A11 and A14 show
590 near-zero precipitation sensitivity, while Cambourne Road, serving new residen-
591 tial developments, is far more vulnerable. The Characterise stage (Section 5.4)
592 identifies the number of lanes and gradient as the strongest observable correlates
593 of precipitation resilience. Crucially, these associations have clear engineering
594 interpretations: multi-lane roads are built to higher DMRB design standards
595 with steeper crossfall and larger drainage capacity, while road gradient enhances
596 gravitational surface water clearance, particularly relevant in the flat fenland
597 study area. Observable characteristics, however, capture only part of the cross-
598 link variation; unobserved factors such as drainage system condition, pavement
599 porosity, and maintenance history may partially explain the residual hetero-
600 geneity.

601 The estimated coefficients are therefore not pure meteorological quantities
602 but *hydro-infrastructure* metrics that capture the realised compound effect of
603 rainfall intensity, drainage design, ageing, and maintenance on each segment’s
604 capacity to manage surface water. This reframing implies that drainage invest-
605 ment prioritisation can move beyond flood history or hydraulic modelling to a
606 network-wide, empirically grounded screening tool: segments identified as highly
607 sensitive are candidates for drainage deficiency, warranting targeted engineering
608 inspection even in the absence of micro-scale drainage data.

609 6.2. Comparison with Prior Literature

610 The baseline estimate (-0.0111 , or 1.1% per mm/hr) sits at the lower end of
611 the literature. The speed-at-capacity reductions of 8–14% reported by Hranac

et al. (2006) capture average effects during sustained rain, not the marginal effect per millimetre. The closest methodological comparator, Becker et al. (2026), used continuous radar precipitation with a log-linear specification and found speed reductions that vary by road type and rainfall intensity, with the present estimate falling within a comparable range. Across the full German road hierarchy, Becker et al. found that multi-lane, high-speed roads experienced proportionally *larger* speed reductions than lower-speed roads ($\Delta v_{\text{rel}} = -26\%$ on multi-lane versus -15% on single-lane roads at 100 km/h), attributing this to the greater headroom for speed reduction on roads with higher baseline speeds. The present study, by contrast, finds that multi-lane segments within the SRN are *less* sensitive per millimetre of rainfall, a finding that the Characterise stage (Section 5.4) links to the higher drainage design standards (crossfall, capacity) mandated for multi-lane roads under the DMRB. The apparent discrepancy likely reflects the difference in scope: Becker et al. compared across the full speed spectrum (50–130 km/h, including urban streets), whereas the present sample is confined to the SRN, where all roads meet a minimum design threshold. Within this narrower range, the drainage-quality advantage of higher-specification roads may dominate the headroom effect that drives Becker et al.’s cross-hierarchy result.

Link-specific estimation reveals that within-network variation in precipitation sensitivity is at least as large as the between-study variation compiled by Koetse and Rietveld (2009). For adaptation planning, the spatial distribution of sensitivity within a network matters more than the precision of any single aggregate estimate.

Gao et al. (2024) also found heterogeneity in rainfall response across Harbin road segments, confirming that the phenomenon is not country-specific. The regression framework presented here, however, isolates the marginal precipitation effect through volume, temporal, and disruption controls that a probabilistic reliability model cannot separate. Continuous coefficients (percentage speed reduction per millimetre) translate directly into infrastructure investment metrics; threshold-based resilience scores do not.

6.3. The Open-Source Platform

The framework is deployed as an open-source Streamlit platform. Users can explore an interactive map of link-level sensitivity coefficients with colour-coded overlays, view corridor-level summary statistics and box-plot distributions, and upload their own WebTRIS and weather data to replicate the analysis for any UK road network with sensor coverage. The platform runs entirely on the user's machine, requiring no cloud infrastructure and respecting local data governance constraints.

6.4. Policy Implications

The most immediate policy use is *prioritising drainage investment*: surface upgrades and capacity enhancements can be directed at the specific segments identified as most hydrologically vulnerable, rather than distributed uniformly or allocated on flood history alone. Applying the scenario loss formula (Equation 3) to a 10 mm/hr storm scenario, Cambourne Road segments would experience an estimated speed reduction of approximately 33%, while A11 segments would lose approximately 7%, a fivefold difference that is invisible in network-average metrics but critical for operationalising climate resilience at the corridor level (Fig. 7).

Beyond targeting, the near-zero sensitivity of the most resilient corridors (A11, A1(M)) implies that higher design standards, particularly the drainage capacity specified in the DMRB for motorways and high-speed dual carriageways, may themselves constitute an effective form of climate resilience, a finding relevant to debates about investment standards for new construction (National Highways, 2024). The open-source platform gives local authorities the evidence base needed to inform adaptation planning and prioritise climate resilience funding applications, such as those administered through the DARE network.

For example, the two Cambourne Road links (with coefficients of -0.039) serve a rapidly growing population catchment. The Characterise stage (Section 5.4) suggests that their single-carriageway design, with lower crossfall gradient and smaller drainage capacity than dual-carriageway alternatives, may

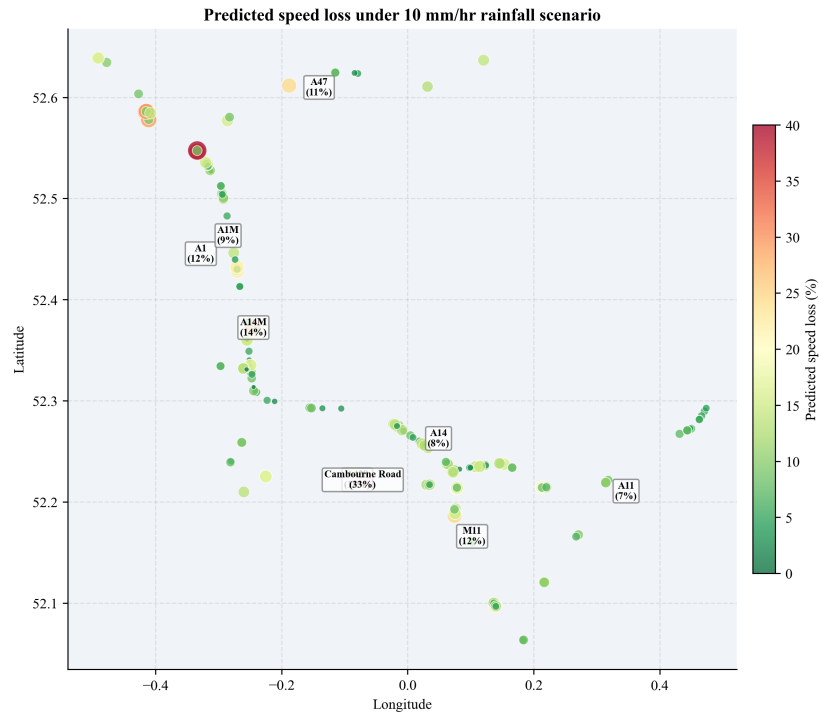


Figure 7: Predicted speed loss under a 10 mm/hr rainfall scenario. Colour scale ranges from green (minimal loss, <10%) to red (severe loss, >25%). The map translates per-millimetre coefficients into absolute scenario impacts that are directly interpretable by transport planners and drainage engineers.

partially explain their extreme sensitivity; if so, targeted drainage upgrades could meaningfully reduce the current speed penalty during moderate rainfall. At the corridor level, the A1’s extreme within-corridor variation (s.d. = 0.013) identifies it as a priority for segment-by-segment engineering assessment to distinguish its most resilient from its most vulnerable links.

The scenario loss formula (Equation 3) also provides a natural bridge to forward-looking climate risk assessment. By substituting the rainfall intensity R with design storm values projected under future climate scenarios, such as UKCP18 high-emission projections for the 2080s (Department for Transport, 2024), planners can estimate how link-level speed losses may evolve as extreme rainfall events become more frequent and intense. The open-source platform already accepts user-specified rainfall intensities, enabling this extension without re-estimation of the underlying model. This positions the tool not only as a diagnostic instrument for current vulnerability but also as a screening tool for future climate adaptation priorities.

6.5. Robustness Checks

Four checks assess the sensitivity of the main results to modelling choices.

Excluding traffic volume. Because speed and volume are jointly determined (Section 4.4), including volume as a control could absorb part of the rainfall effect that operates through reduced demand. Re-estimating the Gamma GLM without the volume variable yields precipitation coefficients that preserve the same sign and corridor-level ranking. The baseline coefficient shifts from -0.0111 to approximately -0.0106 , a 4.3% attenuation consistent with the demand-reduction channel: when volume is omitted, part of the speed recovery associated with fewer vehicles during rain is attributed to the precipitation variable, pulling the coefficient toward zero. The corridor-level pattern (A11 most resilient, Cambourne Road most sensitive) is unchanged.

Peak hours only. Restricting the estimation sample to weekday peak hours (07:00–09:00 and 16:00–19:00) tests whether the results are driven by off-peak observations with atypical speed–weather relationships. The peak-hour baseline

precipitation coefficient (-0.0181) is larger in magnitude than the full-sample estimate (-0.0111), consistent with precipitation having a stronger marginal effect when traffic is already near capacity. The rank ordering of corridors by sensitivity is preserved, confirming that the link-level heterogeneity pattern is not an artefact of time-of-day composition.

Alternative distributional specification. As noted in Section 4.2, an OLS model on untransformed speed was estimated as an alternative. The OLS precipitation coefficient (-0.559 mph per mm/hr) confirms the direction and statistical significance of the Gamma GLM estimate. At the sample mean speed of 54.3 mph, this implies a 1.0% reduction per mm/hr, consistent with the Gamma GLM’s 1.1% estimate. The OLS specification achieves $R^2 = 0.696$ but imposes a constant absolute (rather than proportional) rainfall effect and does not accommodate the mean–variance relationship in the speed data.

Extreme rainfall subsample. To test whether the estimated sensitivities are amplified during intense precipitation, the scenario most relevant to climate adaptation, the model was re-estimated on the subsample of hours with precipitation exceeding 5 mm/hr, the UK Met Office threshold for heavy rainfall. Only 68,679 weekday hours (1.3% of the full sample) exceed this threshold; after the GLM drops observations with sparse link–hour cells due to collinearity, the estimation sample is $N = 7,453$. The reference-link baseline coefficient increases in magnitude from -0.011 to -0.129 ($SE = 0.002$, $z = -74.3$), an order-of-magnitude amplification consistent with the physical expectation that drainage systems approach capacity during heavy rainfall, producing a non-linear increase in surface water depth. The link-specific interaction terms shift proportionally, so the cross-link ranking structure is qualitatively preserved even as absolute magnitudes change. Because the extreme subsample retains only 1.3% of weekday observations, these estimates are necessarily less precise at the individual-link level; nonetheless, the direction and scale of the amplification confirm that the hydrological vulnerability identified in the full sample is *further exacerbated* under the extreme rainfall scenarios projected by UK climate models.

Table 5 summarises the control variable coefficients across specifications. Ta-

ble 6 reports the corridor-level mean precipitation sensitivity under each Gamma GLM specification. The corridor ranking is identical across all three specifications: Cambourne Road is the most sensitive and A11 the most resilient regardless of whether volume is included or the sample is restricted to peak hours. The core finding, that link-level precipitation sensitivity varies by a factor of approximately 20 and clusters by corridor with clear spatial patterns, is robust.

Table 5: Model comparison: Gamma GLM under alternative specifications

	(1)	(2)	(3)	(4)
	Full model	No volume	Peak hours	OLS
Precipitation	−0.0111*** (0.0002)	−0.0106*** (0.0002)	−0.0181*** (0.0002)	−0.5586*** (0.0089)
Volume	−0.00002*** (0.000008)	—	−0.00003*** (0.000006)	−0.0007** (0.0003)
Temperature	0.00032*** (0.0001)	0.00025** (0.0001)	0.00195*** (0.0003)	0.0248*** (0.0054)
School term	−0.0051*** (0.0009)	−0.0051*** (0.0009)	−0.0050*** (0.0012)	−0.3116*** (0.0529)
Event	−0.0236*** (0.0029)	−0.0253*** (0.0030)	−0.0254*** (0.0034)	−1.3872*** (0.1617)
Roadworks	−0.0101*** (0.0023)	−0.0102*** (0.0023)	−0.0055** (0.0024)	−0.6194*** (0.1373)
Accident	−0.0127*** (0.0019)	−0.0130*** (0.0019)	−0.0104*** (0.0021)	−0.7322*** (0.1081)
Link FE	Yes	Yes	Yes	Yes
Link × Precip. interactions	Yes	Yes	Yes	Yes
Temporal FE	Yes	Yes	Yes	Yes
Calendar controls	Yes	Yes	Yes	Yes
Family / link	Gamma / log	Gamma / log	Gamma / log	Gaussian / identity
N	5,050,333	5,050,333	1,475,034	5,050,333
AIC	10.111	10.111	10.148	—
R^2	—	—	—	0.696

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered SE at link level. All models include link FE and link × precip. interactions (201 coefficients). (2) drops volume; (3) peak hours only; (4) OLS. Full link-level coefficients in Appendix A.

Table 6: Corridor-level mean precipitation sensitivity under alternative specifications

Corridor	N	(1) Full	(2) No volume	(3) Peak hours	Rank changed?
Cambourne Rd	2	-0.0392	-0.0387	-0.0462	No
A14(M)	8	-0.0149	-0.0144	-0.0219	No
A1	30	-0.0137	-0.0132	-0.0207	No
M11	29	-0.0124	-0.0119	-0.0194	No
A47	10	-0.0123	-0.0118	-0.0193	No
A428	13	-0.0122	-0.0117	-0.0192	No
A1(M)	28	-0.0095	-0.0091	-0.0166	No
A14	72	-0.0087	-0.0082	-0.0157	No
A11	13	-0.0075	-0.0070	-0.0145	No

Mean total precipitation sensitivity ($\beta_p + \bar{\delta}_i$) by corridor. Corridors sorted from most sensitive (Cambourne Rd) to most resilient (A11). Ranking is identical across all three Gamma GLM specifications.

6.6. Limitations

The study area is limited to Cambridgeshire and Peterborough; while the methodology transfers, the specific coefficients reflect local conditions, particularly the flat, low-lying fenland terrain, and should not be applied directly elsewhere. Although traffic volume is included as a control to account for the speed-flow relationship, the Gamma GLM estimates a static hourly effect and does not model the temporal dynamics of how congestion builds and dissipates during a storm event (e.g., queue formation and clearance). This is by design: the static specification deliberately isolates the supply-side infrastructure penalty from demand-side congestion dynamics, which is the parameter most relevant for drainage investment prioritisation.

A fundamental data limitation is that “precipitation” as measured at weather stations is a proxy variable: the quantity that directly affects driving conditions is the depth of surface water on the road, which depends on the interaction between rainfall intensity and the segment’s drainage capacity, pavement per-

meability, and crossfall geometry. The estimated coefficients therefore capture this compound effect rather than a pure meteorological response. While this limits its mechanistic decomposition, it enhances practical relevance: the coefficients directly identify the segments where the rainfall–drainage interaction produces the largest speed penalties, which is precisely the information needed to prioritise adaptation investment.

On the data side, nearest-station weather matching may introduce measurement error for links far from monitoring stations; radar-based precipitation data would improve spatial precision, though the resulting measurement error likely attenuates the coefficients towards zero, making the reported sensitivities conservative lower bounds of the true meteorological effect.

A deliberate methodological boundary of this study is the exclusion of direct tailpipe emission estimation. Translating link-level speed penalties into emission increments requires granular, time-varying data on fleet composition, vehicle age, and fuel types—parameters that are not publicly available at the segment level. Introducing unverified assumptions regarding these fleet characteristics would risk contaminating the empirically grounded speed estimates with structural emission-modelling errors. This study therefore provides the most spatially disaggregated first-order effect (the supply-side speed penalty) as a foundational input that local authorities can integrate with proprietary fleet data in downstream models such as COPERT or HBEFA when emission quantification is required.

A further consideration is spatial spillover within the road network. When surface water accumulation forces lane closures or speed reductions on a given segment, traffic may divert to parallel routes, increasing volumes and potentially degrading speeds on those alternative links. This mechanism could attenuate the estimated precipitation coefficient on the affected segment (fewer vehicles observed during heavy rain) while inflating the coefficient on receiving segments (increased congestion misattributed to weather). However, because the model conditions on observed hourly traffic volume, the precipitation coefficient captures the speed effect of rainfall at any given level of demand. Provided that

786 rainfall-induced diversion scales approximately in proportion to rainfall inten-
787 sity across links, the relative ranking of link-level sensitivity is unlikely to be
788 systematically distorted. Formally incorporating spatial dependence through a
789 spatial lag or spatial error model would require defining a network adjacency
790 matrix and is deferred to future work.

791 *6.7. Future Research*

792 The immediate next step is to apply the framework to the local road network
793 in Cambridgeshire and Peterborough using County Council data, comparing
794 SRN and local road sensitivity and extending coverage to the road classes that
795 carry most daily trips. Extending the framework to other weather variables
796 (wind, fog, snow) that affect the SRN would broaden the climate risk pro-
797 file beyond precipitation. A complementary extension is network vulnerability
798 analysis: betweenness centrality under normal and extreme-weather scenarios
799 would identify links whose rainfall degradation cascades into network-wide jour-
800 ney time losses, connecting these link-level estimates to system-level climate risk
801 assessment (Ganin et al., 2017).

802 The link-level speed losses estimated here can also serve as inputs to down-
803 stream environmental impact models. Speed-dependent emission factors from
804 established frameworks such as COPERT or HBEFA, combined with the pre-
805 dicted speed reductions under rainfall scenarios, would allow estimation of the
806 additional CO₂ and pollutant emissions attributable to precipitation-induced
807 congestion on each segment. This analytical chain, from spatial vulnerability
808 identification to emissions quantification, would connect the present framework
809 to the broader transport–environment policy agenda, though constructing such
810 coupled models is beyond the scope of this paper.

811 **7. Conclusion**

812 This paper posed three questions about the hydrological vulnerability of the
813 UK Strategic Road Network to precipitation. The answers are clear: individual

814 link-level sensitivity varies by a factor of approximately 20 (corridor averages
815 by a factor of 5) within a single region, with corridor-level spatial patterns
816 that distinguish resilient from vulnerable segments; observable road character-
817 istics, particularly lane count and gradient (which proxy for drainage design
818 standards), explain a meaningful share of the cross-link heterogeneity; and pro-
819 ducing this evidence requires only public data and a standard laptop. Together,
820 these findings demonstrate that rainfall vulnerability is not a uniform meteoro-
821 logical exposure but a heterogeneous property shaped by design standards and
822 physical geometry.

823 Three broader implications follow for transport–environment policy. First,
824 network-average vulnerability metrics, which currently inform most adaptation
825 budgets, are too coarse for efficient investment allocation. Link-level coefficients
826 make it possible to prioritise drainage investment at the segments where each
827 pound spent yields the largest reduction in rainfall-induced delay. Second, the
828 Characterise stage demonstrates that observable road attributes, particularly
829 lane count and gradient (which proxy for drainage design standards), explain a
830 substantial share of the heterogeneity, providing testable hypotheses for targeted
831 engineering inspection of high-sensitivity segments. Third, the methodological
832 barrier to link-level analysis is lower than the dominance of simulation-based
833 approaches suggests. A Gamma GLM with road link fixed effects and inter-
834 action terms is sufficient to recover interpretable sensitivity estimates without
835 microsimulation or high-performance computing, putting evidence generation
836 within reach of local authorities who lack specialised modelling resources. The
837 coefficients themselves do not simulate the effect of specific interventions (e.g.,
838 drainage upgrades); rather, they provide the spatially disaggregated evidence
839 base on which such investment decisions can be grounded, helping to opera-
840 tionalise climate resilience at the local authority level.

841 The open-source platform and codebase are released to enable replication
842 across any UK road network with WebTRIS coverage.

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848 Archive.

849 **CRedit authorship contribution statement**

850 **Yibin Li:** Conceptualization, Methodology, Software, Formal analysis, Data
851 curation, Writing – Original Draft, Visualization. **Li Wan:** Conceptualization,
852 Supervision, Writing – Review & Editing, Funding acquisition, Project admin-
853 istration.

854 **Declaration of competing interest**

855 The authors declare no competing financial interests or personal relation-
856 ships that could have influenced this work.

857 **Data availability**

858 All data are publicly available. Traffic data: National Highways WebTRIS
859 (<https://webtris.nationalhighways.co.uk/>). Weather data: Met Office
860 CEDA Archive (<https://data.ceda.ac.uk/>). Code and platform: [GitHub
861 URL upon publication].

862 **AI Disclosure**

863 AI-assisted tools were used for literature search, coding assistance, and draft-
864 ing support. All content was reviewed, verified, and approved by the authors,
865 who accept full responsibility for the work.

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Appendix A. Road Link Precipitation Sensitivity: Robustness Across Specifications

Table A.7 reports the total precipitation sensitivity coefficient ($\beta_p + \delta_i$) for all 201 road links under four model specifications. Links are sorted from most

950 sensitive (most negative) to most resilient (least negative).

Table A.7: Road link precipitation sensitivity coefficients under four specifications ($N = 201$)

Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
173	-0.0418***	-0.0421***	-0.0752***	-2.2538***
60	-0.0287***	-0.0284***	-0.0342***	-1.7414***
32	-0.0283***	-0.0277***	0.0109***	-1.1439***
66	-0.0220***	-0.0217***	-0.0205***	-1.4262***
62	-0.0201***	-0.0196***	-0.0162***	-1.2416***
145	-0.0199***	-0.0198***	-0.0145***	-1.1675***
136	-0.0198***	-0.0196***	-0.0291***	-1.2306***
124	-0.0183***	-0.0180***	-0.0222***	-1.1619***
206	-0.0176***	-0.0172***	-0.0162***	-0.8251***
121	-0.0172***	-0.0169***	-0.0103***	-0.8981***
30	-0.0172***	-0.0170***	-0.0117***	-0.7999***
192	-0.0164***	-0.0162***	-0.0119***	-0.9061***
198	-0.0164***	-0.0163***	-0.0134***	-0.8576***
213	-0.0161***	-0.0162***	-0.0166***	-0.9612***
199	-0.0156***	-0.0154***	-0.0122***	-0.5783***
147	-0.0154***	-0.0152***	-0.0093***	-0.8166***
29	-0.0153***	-0.0151***	-0.0156***	-0.7138***
154	-0.0152***	-0.0150***	-0.0242***	-0.8585***
174	-0.0150***	-0.0148***	-0.0058***	-0.6395***
104	-0.0150***	-0.0147***	-0.0082***	-0.7798***
191	-0.0149***	-0.0147***	-0.0105***	-0.7318***
204	-0.0148***	-0.0144***	-0.0090***	-0.7959***
182	-0.0145***	-0.0147***	-0.0143***	-0.8217***
78	-0.0144***	-0.0144***	-0.0095***	-0.7233***

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Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
77	-0.0143***	-0.0143***	-0.0104***	-0.7112***
155	-0.0143***	-0.0138***	-0.0129***	-0.7618***
150	-0.0141***	-0.0144***	-0.0163***	-0.4201***
88	-0.0141***	-0.0139***	-0.0082***	-0.6710***
164	-0.0141***	-0.0138***	-0.0097***	-0.8141***
123	-0.0139***	-0.0138***	-0.0132***	-0.7138***
3	-0.0139***	-0.0138***	-0.0162***	-0.9070***
208	-0.0139***	-0.0139***	-0.0112***	-0.8440***
116	-0.0138***	-0.0138***	-0.0097***	-0.7060***
26	-0.0137***	-0.0141***	-0.0110***	-0.7554***
4	-0.0136***	-0.0136***	-0.0104***	-0.6703***
138	-0.0136***	-0.0135***	-0.0126***	-0.7618***
14	-0.0136***	-0.0135***	-0.0105***	-0.6185***
61	-0.0132***	-0.0132***	-0.0112***	-0.6451***
163	-0.0130***	-0.0126***	-0.0229***	-0.7450***
144	-0.0129***	-0.0130***	-0.0113***	-0.6746***
193	-0.0129***	-0.0127***	-0.0076***	-0.6265***
129	-0.0129***	-0.0127***	-0.0087***	-0.6627***
89	-0.0129***	-0.0127***	-0.0126***	-0.6998***
91	-0.0126***	-0.0122***	-0.0080***	-0.6945***
93	-0.0125***	-0.0123***	-0.0086***	-0.6370***
196	-0.0125***	-0.0124***	-0.0134***	-0.4337***
51	-0.0123***	-0.0121***	-0.0070***	-0.5847***
118	-0.0122***	-0.0120***	-0.0084***	-0.7589***
1	-0.0122***	-0.0120***	-0.0113***	-0.5515***
133	-0.0121***	-0.0120***	-0.0178***	-0.8409***
63	-0.0118***	-0.0116***	-0.0073***	-0.5248***
20	-0.0118***	-0.0116***	-0.0078***	-0.5651***

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Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
112	−0.0115***	−0.0114***	−0.0098***	−0.7011***
119	−0.0111***	−0.0110***	−0.0070***	−0.6609***
146	−0.0111	−0.0106	−0.0181	−0.5586
120	−0.0110***	−0.0106***	−0.0120***	−0.7200***
212	−0.0110***	−0.0118***	−0.0118***	−0.2250***
194	−0.0108***	−0.0106***	−0.0097***	−0.6973***
211	−0.0106***	−0.0114***	−0.0158***	−0.1977***
64	−0.0106***	−0.0106***	−0.0098***	−0.6589***
137	−0.0105***	−0.0102***	−0.0180***	−0.6956***
187	−0.0105***	−0.0104***	−0.0084***	−0.6773***
68	−0.0104***	−0.0101***	−0.0088***	−0.6504***
165	−0.0104***	−0.0102***	−0.0112***	−0.6614***
122	−0.0104***	−0.0101***	−0.0171***	−0.6151***
131	−0.0102***	−0.0101***	−0.0089***	−0.5614***
178	−0.0102***	−0.0099***	−0.0058***	−0.5995***
189	−0.0102***	−0.0101***	−0.0067***	−0.4913***
169	−0.0101***	−0.0100***	−0.0068***	−0.6468***
101	−0.0100***	−0.0095***	−0.0056***	−0.6241***
49	−0.0099***	−0.0095***	−0.0075***	−0.6137***
156	−0.0098***	−0.0095***	−0.0071***	−0.5875***
108	−0.0097***	−0.0097***	−0.0064***	−0.5817***
142	−0.0095***	−0.0096***	−0.0079***	−0.6029***
113	−0.0094***	−0.0093***	−0.0074***	−0.3895***
36	−0.0094***	−0.0092***	−0.0073***	−0.5987***
210	−0.0092***	−0.0092***	−0.0048***	−0.4465***
33	−0.0091***	−0.0091***	−0.0073***	−0.4622***
50	−0.0091***	−0.0091***	−0.0118***	−0.5039***
97	−0.0091***	−0.0089***	−0.0068***	−0.4408***

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Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
100	−0.0090***	−0.0086***	−0.0059***	−0.5059***
25	−0.0089***	−0.0088***	−0.0060***	−0.5040***
31	−0.0089***	−0.0087***	−0.0037***	−0.2714***
22	−0.0089***	−0.0086***	−0.0071***	−0.5752***
158	−0.0089***	−0.0087***	−0.0142***	−0.4100***
15	−0.0088***	−0.0087***	−0.0075***	−0.4265***
67	−0.0088***	−0.0085***	−0.0057***	−0.5413***
195	−0.0086***	−0.0084***	−0.0022***	−0.3644***
172	−0.0086***	−0.0086***	−0.0034***	−0.5624***
152	−0.0086***	−0.0084***	−0.0127***	−0.4597***
53	−0.0086***	−0.0083***	−0.0102***	−0.5295***
55	−0.0085***	−0.0083***	−0.0115***	−0.5262***
83	−0.0085***	−0.0082***	−0.0066***	−0.5303***
84	−0.0084***	−0.0083***	−0.0059***	−0.5230***
37	−0.0083***	−0.0078***	−0.0064***	−0.5239***
190	−0.0083***	−0.0081***	−0.0036***	−0.3399***
181	−0.0082***	−0.0079***	−0.0078***	−0.4972***
188	−0.0081***	−0.0080***	−0.0073***	−0.5392***
44	−0.0081***	−0.0078***	−0.0053***	−0.4030***
16	−0.0080***	−0.0077***	−0.0073***	−0.5061***
23	−0.0079***	−0.0076***	−0.0125***	−0.4674***
11	−0.0079***	−0.0076***	−0.0070***	−0.4952***
52	−0.0078***	−0.0076***	−0.0099***	−0.4771***
166	−0.0077***	−0.0066***	−0.0188***	−0.4875***
175	−0.0077***	−0.0075***	−0.0042***	−0.3596***
73	−0.0077***	−0.0077***	−0.0082***	−0.3388***
59	−0.0077***	−0.0077***	−0.0096***	−0.2768***
179	−0.0076***	−0.0073***	−0.0063***	−0.4382***

Continued on next page

Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
197	−0.0076***	−0.0074***	−0.0078***	−0.4936***
9	−0.0076***	−0.0075***	−0.0091***	−0.4548***
186	−0.0075***	−0.0074***	−0.0063***	−0.4841***
200	−0.0075***	−0.0073***	0.0105***	−0.2994***
115	−0.0075***	−0.0073***	−0.0059***	−0.4655***
209	−0.0074***	−0.0071***	−0.0069***	−0.4651***
167	−0.0074***	−0.0070***	−0.0038***	−0.4804***
56	−0.0074***	−0.0071***	−0.0105***	−0.4447***
42	−0.0073***	−0.0071***	−0.0066***	−0.4663***
54	−0.0072***	−0.0069***	−0.0051***	−0.3294***
132	−0.0071***	−0.0069***	−0.0056***	−0.3017***
157	−0.0070***	−0.0067***	−0.0064***	−0.3727***
35	−0.0068***	−0.0064***	−0.0048***	−0.2911***
8	−0.0068***	−0.0065***	−0.0086***	−0.4015***
12	−0.0067***	−0.0065***	−0.0073***	−0.3776***
184	−0.0067***	−0.0067***	−0.0046***	−0.4192***
7	−0.0067***	−0.0066***	−0.0078***	−0.4037***
143	−0.0066***	−0.0065***	−0.0056***	−0.2707***
18	−0.0064***	−0.0063***	−0.0050***	−0.2731***
82	−0.0064***	−0.0061***	−0.0048***	−0.3822***
87	−0.0064***	−0.0062***	−0.0040***	−0.2717***
135	−0.0064***	−0.0060***	−0.0052***	−0.3825***
148	−0.0061***	−0.0060***	−0.0035***	−0.3905***
168	−0.0061***	−0.0057***	−0.0063***	−0.3890***
75	−0.0060***	−0.0059***	−0.0045***	−0.2223***
65	−0.0059***	−0.0030***	−0.0007	−0.1223***
203	−0.0059***	−0.0057***	−0.0011***	−0.2487***
117	−0.0059***	−0.0058***	−0.0036***	−0.1524***

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Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
183	-0.0058***	-0.0057***	-0.0042***	-0.3464***
2	-0.0058***	-0.0055***	-0.0117***	-0.3581***
28	-0.0057***	-0.0058***	-0.0120***	-0.1812***
39	-0.0056***	-0.0054***	-0.0047***	-0.3435***
103	-0.0056***	-0.0051***	-0.0034***	-0.3275***
153	-0.0055***	-0.0054***	-0.0173***	-0.3348***
69	-0.0055***	-0.0052***	-0.0055***	-0.3332***
95	-0.0055***	-0.0054***	-0.0044***	-0.2818***
149	-0.0054***	-0.0052***	-0.0061***	-0.1684***
71	-0.0053***	-0.0051***	-0.0074***	-0.3002***
47	-0.0053***	-0.0049***	-0.0050***	-0.3193***
45	-0.0051***	-0.0048***	-0.0069***	-0.3072***
72	-0.0051***	-0.0051***	-0.0036***	-0.3809***
96	-0.0051***	-0.0049***	-0.0040***	-0.2040***
10	-0.0050***	-0.0044***	-0.0055***	-0.2824***
110	-0.0050***	-0.0048***	-0.0033***	-0.2274***
127	-0.0049***	-0.0046***	-0.0064***	-0.3273***
24	-0.0049***	-0.0047***	-0.0050***	-0.2924***
99	-0.0049***	-0.0044***	-0.0039***	-0.2885***
76	-0.0047***	-0.0048***	-0.0057***	-0.3248***
48	-0.0047***	-0.0048***	-0.0050***	-0.3017***
92	-0.0046***	-0.0045***	-0.0041***	-0.2985***
177	-0.0045***	-0.0053***	-0.0027***	-0.2644***
41	-0.0044***	-0.0039***	-0.0056***	-0.2658***
105	-0.0041***	-0.0038***	-0.0017***	-0.2413***
5	-0.0041***	-0.0036***	-0.0055***	-0.2430***
46	-0.0037***	-0.0037***	-0.0036***	-0.1889***
134	-0.0036***	-0.0036***	-0.0074***	-0.2345***

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Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
98	−0.0036***	−0.0031***	−0.0080***	−0.2113***
126	−0.0034***	−0.0032***	−0.0040***	−0.2096***
202	−0.0033***	−0.0032***	−0.0028***	−0.2040***
80	−0.0031***	−0.0027***	−0.0040***	−0.1791***
40	−0.0030***	−0.0029***	−0.0028***	−0.1022***
74	−0.0030***	−0.0027***	−0.0046***	−0.1975***
106	−0.0027***	−0.0023***	−0.0046***	−0.1554***
34	−0.0025***	−0.0022***	−0.0025***	−0.0770***
58	−0.0024***	−0.0020***	−0.0042***	−0.1457***
38	−0.0023***	−0.0022***	−0.0031***	−0.0927***
17	−0.0022***	−0.0018***	−0.0031***	−0.1221***
107	−0.0021***	−0.0019***	−0.0031***	−0.1285***
27	−0.0018***	−0.0016***	−0.0116***	−0.1189***
57	−0.0017***	−0.0016***	−0.0051***	−0.0767***
128	−0.0016***	−0.0012***	−0.0024***	−0.0300***
6	−0.0014***	−0.0011***	−0.0036***	−0.0791***
125	−0.0013***	−0.0009***	−0.0036***	−0.0771***
102	−0.0012***	−0.0010***	−0.0010***	−0.0666***
162	−0.0006**	−0.0004	−0.0243***	−0.0649***
114	−0.0006***	−0.0007***	−0.0035***	−0.0322***
140	−0.0006***	−0.0005***	−0.0029***	−0.0331***
109	−0.0004***	−0.0002*	−0.0004**	−0.0141**
19	−0.0003**	0.0002	−0.0029***	−0.0027
13	−0.0003**	−0.0000	−0.0015***	−0.0144**
86	−0.0002	0.0000	−0.0017***	0.0001
159	0.0004	0.0017***	−0.0078***	0.0622**
79	0.0012***	0.0014***	−0.0007***	0.0785***
21	0.0015***	0.0016***	−0.0057***	0.0886***

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Link	(1) Full GLM	(2) No volume	(3) Peak hours	(4) OLS
90	0.0020***	0.0023***	−0.0003**	0.1361***
81	0.0030***	0.0031***	0.0001	0.1975***
161	0.0042***	0.0042***	−0.0048***	0.0235***
207	0.0070***	0.0072***	0.0197***	0.3363***
111	0.0078***	0.0084***	0.0065***	0.4942***
94	0.0086***	0.0088***	0.0039***	0.5675***
139	0.0152***	0.0154***	0.0117***	0.8995***
160	0.0165***	0.0160***	−0.0005	0.4628***
151	0.0188***	0.0188***	−0.0056***	0.3830***

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Total coefficient = baseline + link interaction.

Sorted from most sensitive to most resilient. OLS coefficients in mph per mm/hr.

951 Appendix B. Model Diagnostics

952 Fig. B.8 presents three diagnostic checks for the Gamma GLM. Panel (a)
953 plots deviance residuals against fitted values; the binned mean (orange line) is
954 flat near zero across the full range of predicted speeds, indicating no system-
955 atic misspecification. Panel (b) shows a Q–Q plot of deviance residuals against
956 a normal distribution; the central quantiles align closely with the 45-degree
957 reference, with modest departures in the tails reflecting extreme-weather ob-
958 servations. Panel (c) plots observed against predicted speed; the correlation is
959 0.85.

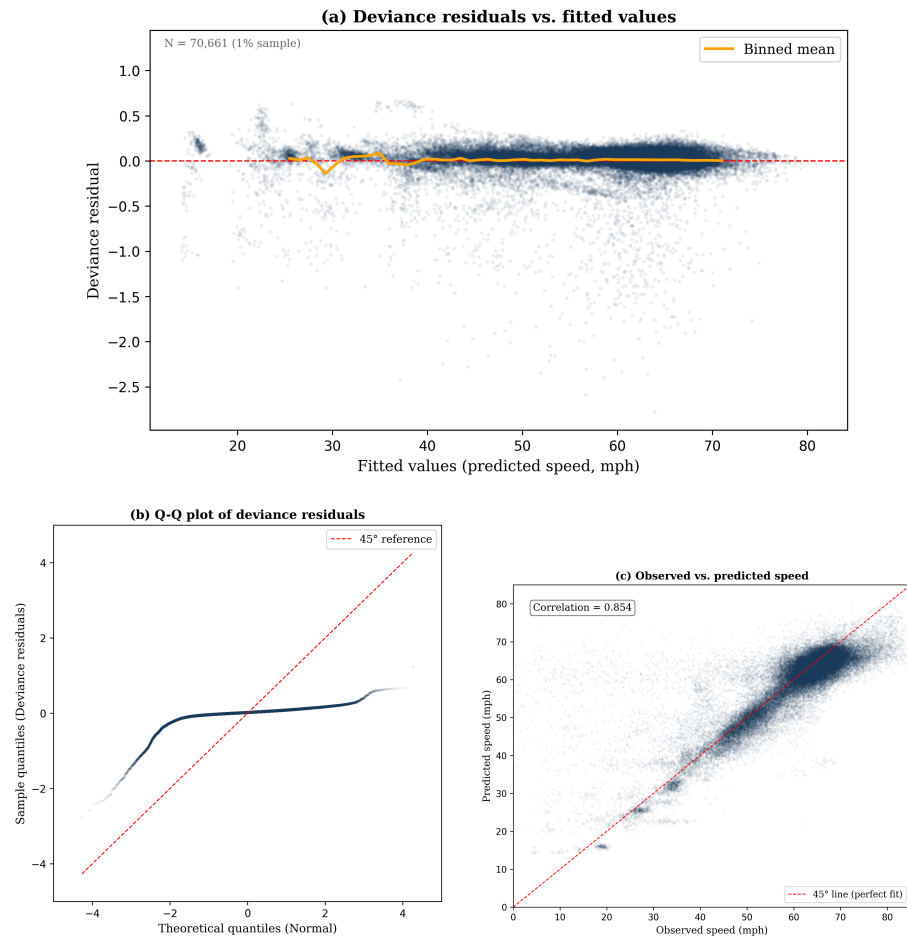


Figure B.8: Model diagnostics for the Gamma GLM (1% random sample, $N = 70,661$). (a) Deviance residuals vs. fitted values with binned mean. (b) Q-Q plot of deviance residuals. (c) Observed vs. predicted speed (correlation = 0.85).